Plasma Sources Science and Technology

PAPER

Coulomb collisions in the Boltzmann equation for electrons in low-temperature gas discharge plasmas

IOP Publishing Plasma Sources Science and Technology
Plasma Sources Sci. Technol. **25** (2016) 015015 (15pp) doi:10.1088/0963-0252/25/1/015015

Coulomb collisions in the Boltzmann equation for electrons in low-temperature gas discharge plasmas

G J M Hagelaar

Université de Toulouse, UPS, INPT, LAPLACE (Laboratoire Plasma et Conversion d'Energie), 118 route de Narbonne, F-31062 Toulouse cedex 9, France
CNRS, LAPLACE, F-31062 Toulouse, France

E-mail: gerjan.hagelaar@laplace.univ-tlse.fr

Received 22 July 2015, revised 2 October 2015
Accepted for publication 28 October 2015
Published 17 December 2015

Abstract

This paper investigates the effects of electron-electron and electron-ion Coulomb collisions on the electron distribution function and transport coefficients obtained from the Boltzmann equation for simple dc gas discharge conditions. Expressions are provided for the full Coulomb collision terms acting on both the isotropic and anisotropic parts of the electron distribution function, which are then incorporated in the freeware Boltzmann equation solver BOLSIG+ . Different Coulomb collision effects are demonstrated and discussed on the basis of BOLSIG+ results for argon gas. It is shown that the anisotropic part of the electron-electron collision term, neglected in previous work, can in certain cases have a large effect on the electron mobility and is essential when describing the transition towards the Coulomb-collision dominated regime characterized by Spitzer transport coefficients. Finally, a brief overview is presented of the discharge conditions for which different Coulomb collision effects occur in different gases.

Keywords: Coulomb collisions, gas discharge, low-temperature plasma, Boltzmann equation, electron transport coefficients, Spitzer conductivity, BOLSIG+

(Some figures may appear in colour only in the online journal)

1. Introduction

Electrons in gas discharges are often far from thermal equilibrium and their distribution function can be far from Maxwellian due to the combined actions of the applied electric field and collisions with the much colder background gas, often near room temperature. It is common practice in low-temperature plasma research to calculate the electron energy distribution function and related electron transport coefficients by solving the Boltzmann equation for a simplified system in which the electric field and the background gas are constant in space and time and in which there are no boundaries. Numerical codes for this type of calculation, so-called electron Boltzmann solvers, were originally developed [1, 2] to analyze data from swarm experiments [3] and are at present widely used to calculate input data for fluid models of gas discharge plasmas [4–6].

Coulomb collisions are usually neglected in these Boltzmann calculations; it is assumed that the ionization degree (defined as the ratio of the electron to neutral particle number densities) is so low that they are completely dominated by electron-neutral collisions. While this assumption is well justified for swarm experiments or Townsend discharges, it is questionable for some other types of discharges, such as positive columns or inductive discharges, in which ionization degrees of 10^{-4} or higher are not rare. One of the first to investigate this question in detail, after preliminary work by Dreicer [7], was Rockwood [8], who incorporated electron-electron collisions in numerical Boltzmann calculations of a dc mercury discharge and found that these collisions cause significant changes in the electron energy distribution function for ionization degrees as low as 3×10^{-5}. This approach was later followed by other works demonstrating Coulomb collision effects at even lower ionization degrees [9–12]. Some of the Boltzmann solvers used in low-temperature plasma modeling, such as ELENDIF [4] or BOLSIG+ [5], include the electron-electron collision term as given by Rockwood.

However, the electron-electron collision term used in these works is incomplete: it only describes how electron-electron energy exchanges influence the isotropic part of the electron distribution function, but does not account for the effect of electron-electron collisions on the anisotropy of the distribution function. It is in fact well known that this latter effect plays a key role in transport phenomena in fully ionized plasmas, where it acts in combination with electron-ion momentum transfer. The theory for this is well developed, starting with the work of Spitzer and co-workers [13, 14], and described in the general plasma physics literature [15–18], but, to our knowledge, it has not been applied to low-temperature plasmas in gas discharges. The low-temperature plasma literature mostly ignores electron-ion collisions and does not make the connection with the well-established Spitzer transport coefficients. On the other hand, Donko [19, 20] recently developed an advanced particle simulation method to describe electron-electron collision effects under swarm conditions, based on Molecular Dynamics and Monte Carlo techniques, which circumvents all collision approximations for the Coulomb interaction and is therefore in principle more accurate than Boltzmann calculations, though at much greater computational costs so that it seems less practical for most low-temperature plasma research purposes.

In this paper, we extend Rockwood's approach and provide expressions for the full Coulomb collision terms including anisotropy effects, both for electron-electron collisions and electron-ion collisions, in a form suitable for inclusion in standard electron Boltzmann solvers for low-temperature plasmas (section 2). We then incorporate these collision terms in the Boltzmann solver BOLSIG+ and demonstrate their influence on the results, on the basis of simple test cases for argon gas (section 3). We show in particular that the anisotropic part of the electron-electron collision term can have a large influence on the electron mobility and is essential when describing the transition towards a Coulomb-collision dominated regime, characterized by Spitzer transport coefficients (sections 3.4 and 3.5). We finally present a brief overview of the discharge conditions for which different Coulomb collision effects occur in different gases (section 3.6) and draw general conclusions (section 4).

Thus, the purpose of this paper is not to improve the fundamental theory of the Coulomb interaction in plasmas, but rather to apply the traditional Coulomb collision theory to low-temperature plasma conditions in way that is more complete than what was previously done in this area. As the work described here has been integrated in the freeware version of BOLSIG+ , available for public download [5], this paper is also intended as documentation and a guide for BOLSIG+ users.

2. Theory

2.1. Basic equations

In this section we introduce the main variables and equations that are typically used to calculate the distribution function and transport coefficients of electrons in gas discharges. The basis of these calculations is the Boltzmann equation for electrons

$$\frac{\partial f}{\partial t} + \mathbf{v}\cdot\nabla f - \frac{e}{m_\mathrm{e}}(\mathbf{E}+\mathbf{v}\times\mathbf{B})\cdot\nabla_\mathbf{v} f = \left(\frac{\partial f}{\partial t}\right)_\mathrm{col}, \quad (1)$$

where f is the electron distribution function in phase space, $\mathbf{v}$ is the velocity vector, $\nabla_\mathbf{v}$ is the gradient operator for velocity space, e is the elementary charge, m_e is the electron mass, $\mathbf{E}$ and $\mathbf{B}$ are the macroscopic electric and magnetic fields, and the right hand side represents the rate of change of the distribution function due to collisions with the background gas. This equation is then applied to the simple case that the magnetic field is zero, the electric field and background gas are constant in space and time, and there are no boundaries. (This corresponds to swarm experiments.) Assuming that the electron motion is nearly isotropic due to scattering in elastic collisions, the distribution function is approximated by a first-order spherical harmonics expansion in velocity space:

$$f(\mathbf{x},\mathbf{v},t) = f_0(\mathbf{x},v,t) + f_1(\mathbf{x},v,t)\cos\theta, \quad (2)$$

where v is the norm of the velocity vector and θ is the velocity angle with respect to the direction of the electric field. This is known as the 'two-term approximation'. In addition it is assumed that the velocity-dependence of the distribution function is independent of time and space:

$$f_{0,1}(\mathbf{x},v,t) = \frac{n_e(\mathbf{x},t)}{2\pi\gamma^3}F_{0,1}(\varepsilon) \quad (3)$$

where n_e is the electron number density and we choose to change the main variable from electron velocity v to electron energy ε using

$$v = \gamma\varepsilon^{1/2}, \qquad \gamma = (2/m_\mathrm{e})^{1/2}. \quad (4)$$

Note that some papers [5] and text books [21] define the variables for particle energy and temperature as equivalent voltages, replacing $\varepsilon \to e\varepsilon$ and $kT \to eT$, where k is the Boltzmann constant, as well as $\gamma \to \gamma/\mathrm{e}^{1/2}$ so that $\gamma = (2e/m_\mathrm{e})^{1/2}$. We will not do that here and respect the official SI dimensions of energy and temperature. The isotropic part F_0 of the distribution function corresponds to the electron energy distribution function (EEDF), or more precisely to the electron energy probability function (EEPF) [21], and is normalized as

$$\int_0^\infty \varepsilon^{1/2}F_0(\varepsilon)\mathrm{d}\varepsilon = 1. \quad (5)$$

The anisotropic part F_1 carries the electron mean velocity w due to the electric field, also called drift velocity:

$$w = \frac{\gamma}{3}\int_0^\infty \varepsilon F_1 \mathrm{d}\varepsilon. \quad (6)$$

With these approximations, the Boltzmann equation (1) takes the form of two coupled integro-differential equations for F_0 and F_1 [22, 23]:

$$-\frac{\gamma eE}{3\varepsilon^{1/2}}\frac{\partial}{\partial\varepsilon}(\varepsilon F_1) = C_0^{\mathrm{en}} + C_0^{\mathrm{ee}} + C_0^{\mathrm{ei}} \tag{7}$$

$$-\gamma eE\varepsilon^{1/2}\frac{\partial F_0}{\partial\varepsilon} = C_1^{\mathrm{en}} + C_1^{\mathrm{ee}} + C_1^{\mathrm{ei}} \tag{8}$$

The terms $C_{0,1}$ on the right hand side correspond to the rate of change $(\partial F_{0,1}/\partial t)$ due to different kinds of collisions and generally contain derivatives and/or integrals involving F_0 and F_1. For electron-neutral collisions, these terms are well known and described in the literature [22, 23]. The electron-neutral term acting on F_1 is very simple:

$$C_1^{\mathrm{en}} = -N\gamma\varepsilon^{1/2}\sigma_{\mathrm{m}}^{\mathrm{en}}F_1 \tag{9}$$

where N is the neutral gas particle number density and $\sigma_{\mathrm{m}}^{\mathrm{en}}$ is an effective cross-section for momentum transfer, summed over all electron-neutral processes. Therefore, in the absence of Coulomb collisions, equation (8) becomes

$$F_1 = \frac{eE}{N\sigma_{\mathrm{m}}^{\mathrm{en}}}\frac{\partial F_0}{\partial\varepsilon}, \tag{10}$$

which together with equation (7) yields [1, 2]

$$-\frac{\gamma}{3}\left(\frac{eE}{N}\right)^2\frac{\partial}{\partial\varepsilon}\left(\frac{\varepsilon}{\sigma_{\mathrm{m}}^{\mathrm{en}}}\frac{\partial F_0}{\partial\varepsilon}\right) = \frac{1}{N}\varepsilon^{1/2}C_0^{\mathrm{en}}[F_0]. \tag{11}$$

This is a convenient equation for F_0 alone, where electron heating appears on the left hand side as a diffusion term in energy space. The electron-neutral C_0 term on the right hand side consists of separate contributions from different collision processes, typically of the following form:

$$\begin{aligned}C_0^{\mathrm{en}} = &\frac{2m_{\mathrm{e}}}{M}\frac{N\gamma}{\varepsilon^{1/2}}\frac{\partial}{\partial\varepsilon}\left(\varepsilon^2\sigma_{\mathrm{m}}^{\mathrm{en}}\left(F_0 + kT_0\frac{\partial F_0}{\partial\varepsilon}\right)\right)\\ &-\frac{N\gamma}{\varepsilon^{1/2}}\sum_k(\varepsilon\sigma_k(\varepsilon)F_0(\varepsilon) - (\varepsilon+U_k)\sigma_k(\varepsilon+U_k)F_0(\varepsilon+U_k)) + \ldots\end{aligned} \tag{12}$$

The first term of this expression describes elastic energy losses as a continuous electron flow in energy space, depending on the gas particle mass M and temperature T_0. The second term accounts for discrete energy losses in inelastic processes k with cross sections σ_k and threshold energies U_k, removing electrons from the high-energy tail of the distribution function and reinserting them an interval U_k lower.

In equations (7)–(12), for simplicity but without loss of generality, we have disregarded various matters not directly of interest here, such as growth or decay of the electron number density due to ionization or attachment, superelastic inverse processes, anisotropy of inelastic scattering, gas mixture composition fractions, etcetera. For a more complete description we refer to the literature [e.g. 22–24, 5].

In the next sections we describe the Coulomb collision terms and the consequences they entail for the above Boltzmann calculations.

2.2. Derivation of Coulomb collision terms

According to traditional plasma theory, the Coulomb collision terms of the Boltzmann equation can be derived by applying the Fokker-Planck equation to velocity changes due to binary particle-particle interactions obeying an inverse square law of force. This was first done by Chandrasekhar [25] and then improved by Cohen and Spitzer to describe transport phenomena in fully ionized plasmas [13, 14], followed by many other works in plasma physics, in particular in the context of controlled nuclear fusion [15–18, 26]. A very useful formulation of the Fokker-Planck equation for inverse square interactions, based on generalized potentials, was developed by Rosenbluth and co-workers [15, 16]. This formulation allowed Dreicer [7] and Rockwood [8] to obtain the electron-electron collision term C_0^{ee} for equation (7) and assess its effect on electrons in a low-temperature plasma discharge. Below we extend this approach and derive also the other Coulomb collision terms C_1^{ee}, C_0^{ei} and C_1^{ei}.

Our starting point is the Rosenbluth paper [15] which gives a general expression for the collision term for particles of type a interacting with background particles of type b, valid for arbitrary mass ratio $m_{\mathrm{a}}/m_{\mathrm{b}}$ and arbitrary distribution functions f_{a} and f_{b}. For our purposes, particles a are electrons and particles b are either electrons or ions. We assume that both distribution functions f_{a} and f_{b} are of the two-term form (2) and that the anisotropic parts f_1 are small with respect to the isotropic parts f_0. To obtain the Coulomb collision terms for equations (7) and (8), we substitute these two-term distribution functions in Rosenbluth's expression, neglect all second order terms in $f_1 \cdot f_1$, multiply by 1 and $\cos\theta$, respectively, and integrate over angle space. Some additional information on this derivation is given in the appendix; a similar derivation is carried out in the book of Shkarofsky [16]. The final result, in terms of energy distribution functions F_0 and F_1, is

$$C_0^{\mathrm{eb}} = \frac{2n_{\mathrm{b}}Z_{\mathrm{b}}^2\Gamma}{\gamma^3\varepsilon^{1/2}}\frac{\partial}{\partial\varepsilon}\left(\alpha A_0^1F_0 + \frac{2}{3}(A_0^3 + \varepsilon^{3/2}B_0^0)\frac{\partial F_0}{\partial\varepsilon}\right) \tag{13}$$

$$\begin{aligned}C_1^{\mathrm{eb}} = &-\frac{(1+\alpha)n_{\mathrm{b}}Z_{\mathrm{b}}^2\Gamma}{\gamma^3\varepsilon^{3/2}}(A_0^1F_1 - \varepsilon B_1^{-1}F_0)\\ &+\frac{2n_{\mathrm{b}}Z_{\mathrm{b}}^2\Gamma}{\gamma^3\varepsilon}\frac{\partial}{\partial\varepsilon}\left(\alpha\varepsilon^{1/2}A_0^1F_1 + \frac{2}{3}\varepsilon(A_0^3 + \varepsilon^{3/2}B_0^0)\frac{\partial(F_1/\varepsilon^{1/2})}{\partial\varepsilon}\right.\\ &+\frac{1}{3}((2\alpha-1)A_1^2 - (1+\alpha)\varepsilon^{3/2}B_1^{-1})F_0\\ &\left.+\frac{2}{5}(A_1^4 + \varepsilon^{5/2}B_1^{-1})\frac{\partial F_0}{\partial\varepsilon}\right).\end{aligned} \tag{14}$$

Here n_{b} is the number density of the background particles, Z_{b} is their charge number (i.e. an integer number expressing the particle charge in units of elementary charge e),

$$\alpha = m_{\mathrm{e}}/m_{\mathrm{b}} \tag{15}$$

is the electron to background particle mass ratio, A and B are integral functions over the distribution function of the background particles, defined as

$$A_{0,1}^{m}(\varepsilon) = \alpha^{(m-1)/2}\int_{0}^{\varepsilon/\alpha} u^{m/2}F_{0,1}^{\mathrm{b}}(u)\mathrm{d}u$$
$$B_{0,1}^{m}(\varepsilon) = \alpha^{(m-1)/2}\int_{\varepsilon/\alpha}^{\infty} u^{m/2}F_{0,1}^{\mathrm{b}}(u)\mathrm{d}u, \tag{16}$$

and the strength of the Coulomb interaction is characterized by

$$\Gamma = \frac{e^4 \ln\Lambda}{4\pi(\varepsilon_0 m_\mathrm{e})^2}, \quad \Lambda = \frac{12\pi(\varepsilon_0 kT_\mathrm{e})^{3/2}}{e^3 n_\mathrm{e}^{1/2}}, \quad kT_\mathrm{e} \approx \frac{2}{3}\int_0^\infty \varepsilon^{3/2}F_0\mathrm{d}\varepsilon, \tag{17}$$

with ε_0 the vacuum permittivity. The quantity $\ln\Lambda$ is the Coulomb logarithm related to ambipolar screening which cancels the Coulomb interactions at distances larger than the Debye length; Λ is proportional to the number of electrons contained in a Debye sphere. For the low-temperature plasma conditions of interest here, $\ln\Lambda$ is typically in the range 7–15 depending (very weakly) on the electron density and mean energy. The expression of $\ln\Lambda$ given in equation (17) is the standard expression, used in most works, but corrections to this expression exist for rapidly varying fields or cyclotron oscillations (faster than the plasma frequency) as discussed in [16]. Note also that the theory underlying the above equations is accurate only to order $1/\ln\Lambda \approx 10\%$ and gradually breaks down as $\ln\Lambda$ decreases to order unity [14, 16, 27]; the plasma then enters into a 'strongly coupled' regime which requires more advanced theory [28] or particle simulation [19].

In the next sections we develop the Coulomb collision terms (13) and (14) further for the cases of electron-electron collisions and electron-ion collisions.

2.3. Electron-electron collision terms

For electron-electron collisions, the mass ratio $\alpha = 1$ and the background particle distribution function is identical to the electron distribution function. Substituting this in equations (13) and (14) yields:

$$C_0^{\mathrm{ee}} = \frac{2n_\mathrm{e}\Gamma}{\gamma^3\varepsilon^{1/2}}\frac{\partial}{\partial\varepsilon}\left(A_0^1F_0 + \frac{2}{3}(A_0^3 + \varepsilon^{3/2}B_0^0)\frac{\partial F_0}{\partial\varepsilon}\right) \tag{18}$$

$$C_1^{\mathrm{ee}} = \frac{2n_\mathrm{e}\Gamma}{\gamma^3\varepsilon}\frac{\partial}{\partial\varepsilon}\Bigg(A_0^1B_1^{-1} + \varepsilon^{1/2}A_0^1F_1 + \frac{2}{3}\varepsilon(A_0^3 + \varepsilon^{3/2}B_0^0)\frac{\partial(F_1/\varepsilon^{1/2})}{\partial\varepsilon} + \frac{1}{3}(A_1^2 - 2\varepsilon^{3/2}B_1^{-1})F_0 + \frac{2}{5}(A_1^4 + \varepsilon^{5/2}B_1^{-1})\frac{\partial F_0}{\partial\varepsilon}\Bigg) \tag{19}$$

where

$$A_{0,1}^m(\varepsilon) = \int_0^\varepsilon u^{m/2}F_{0,1}(u)\mathrm{d}u, \quad B_{0,1}^m(\varepsilon) = \int_\varepsilon^\infty u^{m/2}F_{0,1}(u)du \tag{20}$$

The collision term given by equation (18) is identical to that derived by Rockwood [7, 8] and cited in many later papers [4, 5, 9–12]. The expression for the second collision term (19) is generally ignored in low-temperature plasma research, but similar expressions, including additional quasilinear approximations, can be found in the general and fusion plasma literature [16, 18].

An essential property of these collision terms is that they conserve energy and momentum, for any shape of the distribution function:

$$\int_0^\infty \varepsilon^{3/2}C_0^{\mathrm{ee}}\mathrm{d}\varepsilon = 0 \tag{21}$$

$$\int_0^\infty \varepsilon C_1^{\mathrm{ee}}\mathrm{d}\varepsilon = 0. \tag{22}$$

In order to make property (22) more apparent, we have rewritten the first term of equation (14) in conservative form in equation (19), using the identity

$$\frac{\partial}{\partial\varepsilon}(A_0^1B_1^{-1}) = \varepsilon^{1/2}B_1^{-1}F_0 - \frac{1}{\varepsilon^{1/2}}A_0^1F_1. \tag{23}$$

Another property of these collision terms is that they vanish for a shifted Maxwellian distribution function, centered at the mean velocity w, with arbitrary electron temperature T_e:

$$F(\varepsilon,\cos\theta) = \frac{2}{\pi^{1/2}(kT_\mathrm{e})^{3/2}}\exp(-[\varepsilon\sin^2\theta + (\varepsilon^{1/2}\cos\theta - w/\gamma)^2]/(kT_\mathrm{e}))$$
$$= F_0^{\mathrm{M}}(\varepsilon) + F_1^{\mathrm{M}}(\varepsilon)\cos\theta + \dots O(m_\mathrm{e}w^2/2kT_\mathrm{e}) \tag{24}$$

where

$$F_0^{\mathrm{M}}(\varepsilon) = \frac{2}{\pi^{1/2}(kT_\mathrm{e})^{3/2}}\exp(-\varepsilon/(kT_\mathrm{e})) \tag{25}$$

$$F_1^{\mathrm{M}}(\varepsilon) = \frac{4w}{\pi^{1/2}\gamma(kT_\mathrm{e})^{5/2}}\varepsilon^{1/2}\exp(-\varepsilon/(kT_\mathrm{e})). \tag{26}$$

It can be readily verified that for these particular shapes of F_0 and F_1, both (18) and (19) are identically zero. This implies that whenever these collision terms prevail over the other terms of the Boltzmann equation, they force the shapes of F_0 and F_1 to be close to the above Maxwellian shapes.

Including the electron-electron collision terms (18) and (19) in the Boltzmann calculations of section 2.1 is complicated by several issues. The first is that these terms are nonlinear via the integrals functions A and B given in equation (20). This can be handled through an iterative numerical procedure, updating A and B explicitly and then solving for $F_{0,1}$ with standard linear methods, using a conservative discretization scheme to ensure the conservation properties (21) and (22) [5, 8]. A second complication is that the C_1^{ee} term (19) is more complex than the electron-neutral C_1^{en} term (9), destroying the local relation (10) between F_1 and $\partial F_0/\partial\varepsilon$, so that F_0 can no longer be isolated in a single equation (11). Hence it is necessary to retain separate equations (7) and (8) and solve F_0 and F_1 by iteration. A practical way to implement such iteration procedure (in an existing Boltzmann code) is to introduce a correction factor in the heating term of equation (11) as follows. First, F_1 is updated from equation (7):

$$\sigma_\mathrm{m}F_1^{j+1} - \frac{1}{\gamma N\varepsilon^{1/2}}C_1^{\mathrm{ee}}[F_1^{j+1}, F_0^j, A^j, B^j] = -\frac{eE}{N}\frac{\partial F_0^j}{\partial\varepsilon}, \tag{27}$$

where the upper indexes j and $j+1$ refer to subsequent iterations and the square brackets indicate schematically how the electron-electron collision term is evaluated. Then F_0 is solved from equation (11) with a correction factor χ to account for the proper shape of F_1:

$$-\frac{\gamma}{3}\left(\frac{eE}{N}\right)^2\frac{\partial}{\partial\varepsilon}\left(\frac{\chi\varepsilon}{\sigma_m}\frac{\partial F_0^{j+1}}{\partial\varepsilon}\right)=\frac{1}{N}\varepsilon^{1/2}\Big(C_0^{ee}[F_0^{j+1},A^j,B^j]$$

$$+C_0^{en+ei}[F_0^{j+1}]\Big),\quad \chi=\frac{N\sigma_m F_1^{j+1}}{E(\partial F_0^j/\partial\varepsilon)}. \tag{28}$$

We incorporated this procedure in the Boltzmann equation solver BOLSIG+ [5], using straightforward (but conservative) discretization schemes, and did not encounter notable difficulties. Results from this BOLSIG+ implementation are presented in section 3.

2.4. Electron-ion collision terms

For electron-ion collisions, the background particles are ions. We assume that they have a Maxwellian distribution function with a given ion temperature and we neglect ion drift:

$$F_0^b(u)=\frac{2}{\pi^{1/2}(kT_i)^{3/2}}\exp(-u/(kT_i)),\quad F_1^b=0. \tag{29}$$

Due to the small mass ratio

$$\alpha=m_e/m_i\ll 1, \tag{30}$$

the integral functions (16) are constant over most of the electron energy range,$\varepsilon\ll\alpha kT_i$:

$$A_0^1=1,\quad A_0^3=\frac{3}{2}\alpha kT_i,\quad B_0^m=A_1^m=B_1^m=0. \tag{31}$$

Substituting this in equations (13) and (14) and approximating for $\alpha\ll 1$, we find the following expressions for the electron-ion Coulomb collision terms, in agreement with [4, 7, 16, 17]:

$$C_0^{ei}=\frac{n_i\gamma}{\varepsilon^{1/2}}\frac{2m_e}{m_i}\frac{\partial}{\partial\varepsilon}\left(\varepsilon^2\sigma_m^{ei}\left(F_0+kT_i\frac{\partial F_0}{\partial\varepsilon}\right)\right) \tag{32}$$

$$C_1^{ei}=-n_i\gamma\varepsilon^{1/2}\sigma_m^{ei}F_1 \tag{33}$$

where

$$\sigma_m^{ei}=\frac{Z_i^2\Gamma}{\gamma^4\varepsilon^2}=\frac{Z_i^2e^4\ln\Lambda}{16\pi\varepsilon_0^2\varepsilon^2} \tag{34}$$

is an effective cross section for momentum transfer in electron-ion collisions. As is to be expected, these collision terms are similar to those for elastic electron-neutral collisions, given in equations (9) and (12). They fit directly into the Boltzmann calculations described in section 2.1, without additional complications.

3. Results and discussion

In the next sections we illustrate the effect of the Coulomb collision terms described above, on the basis of BOLSIG+ calculations. We implemented these terms in the freeware version of BOLSIG+ , available for public download [5], so that all results shown here can be reproduced and extended by the reader.

For a given discharge gas and given set of electron-neutral cross sections, the BOLSIG+ results are uniquely determined by 4 independent parameters:

(1) the reduced electric field E/N, defined as the ratio of the electric field strength to the gas density, expressed in units of 1 Townsend = 1 Td = 10^{-21} Vm^2;
(2) the ionization degree n_e/N, controlling the relative importance of Coulomb collisions with respect to electron-neutral collisions;
(3) the gas temperature T_0, controlling elastic energy transfer at low E/N;
(4) the plasma density n_e (electron density), affecting very weakly the importance of Coulomb collisions through the Coulomb logarithm.

In all calculations shown here, the gas temperature is 300 K and the plasma density in the Coulomb logarithm is 10^{16} m^{-3}, unless otherwise indicated (figure 7).

3.1. Effect of electron-electron collisions on electron energy distribution

We first discuss the effect of electron-electron Coulomb collisions on the electron energy distribution function F_0. We neglect electron-ion collisions for the moment. Figure 1 shows F_0 in argon gas for an ionization degree of 10^{-3} and different reduced fields of 1, 4, 20 and 100 Td. The electron-argon cross section set is taken from the SIGLO database [29] and contains only three cross sections, respectively for momentum transfer, excitation and ionization. These cross sections are also shown in figure 1(a) (dashed lines).

Without Coulomb collisions (red curves), the shape of the energy distribution function F_0 is determined by the shape of the electron-neutral cross sections as a function of energy. Above a certain energy threshold, inelastic collisions generally deplete the distribution function, resulting in a characteristic two-temperature shape. However, in noble gases at low E/N, such as the 1 Td argon case in figure 1(a), the electrons are not able to reach the inelastic thresholds. The distribution function is then fully determined by the equilibrium, locally in energy space, between the electric heating term and the elastic collision term of equation (11), leading to

$$\frac{\partial\ln F_0}{\partial\varepsilon}=-\frac{6m_e}{M}\frac{N}{eE}\varepsilon(\sigma_m^{en})^2. \tag{35}$$

This says that the logarithmic slope of F_0 is directly proportional to the square of the electron-neutral momentum cross section, and is constant only if $\sigma_m^{en}\propto 1/\varepsilon^{1/2}$, which is not in general the case. The momentum cross section for argon has the special feature that it drops almost to zero at low energy, due to the Ramsauer effect, hence F_0 is nearly constant below a few eV.

Inclusion of the Coulomb collision term C_0^{ee} in the calculation (black and blue curves) changes this behavior. Now F_0

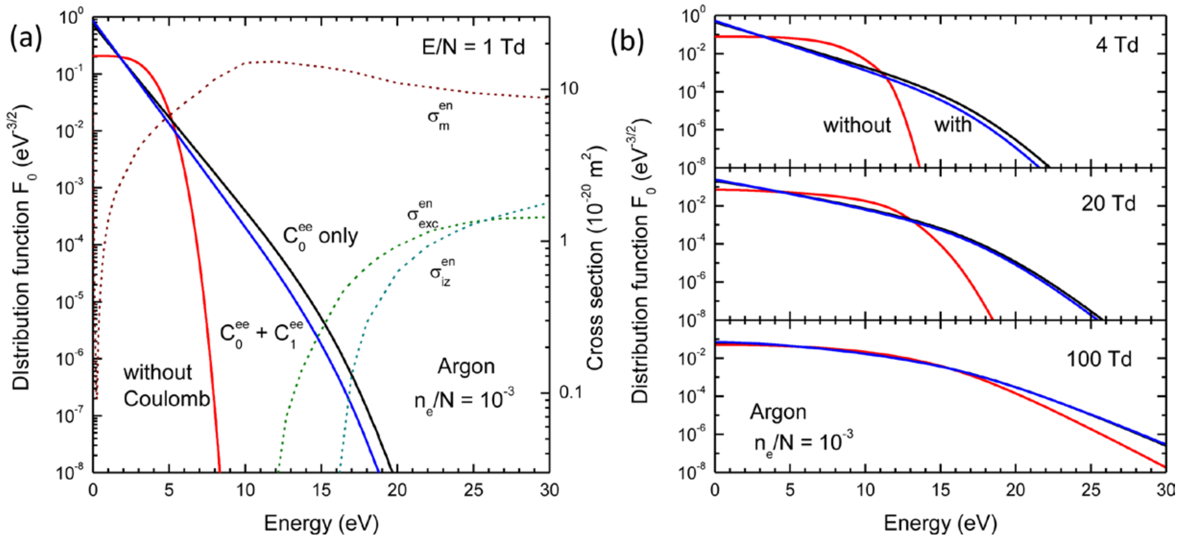

Figure 1. Effect of electron-electron collision terms on the electron distribution function F_0 in argon with an ionization degree of 10^{-3} and different reduced electric fields. The electron-neutral cross sections are also shown (panel (a), dashed lines, right axis). Electron-ion collisions are neglected.

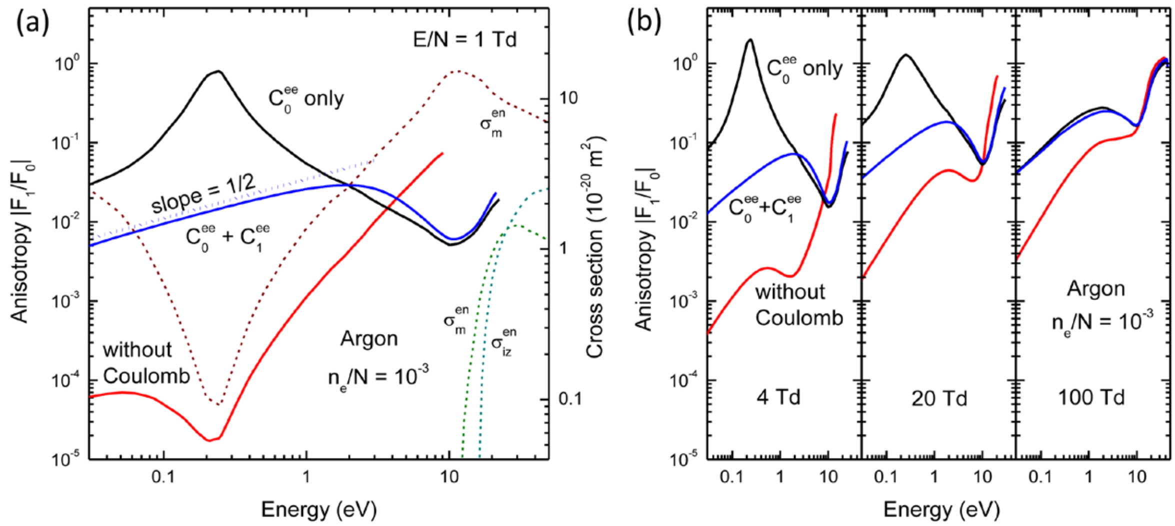

Figure 2. Effect of electron-electron collision terms on the anisotropy $|F_1|/F_0$ in argon for an ionization degree of 10^{-3} and different reduced electric fields. The corresponding distribution functions F_0 are shown in figure 1. The electron-neutral cross sections are also shown (panel (a), dashed lines, right axis). Electron-ion collisions are neglected.

tends to approach the Maxwellian function (25) which has a constant logarithmic slope $-1/(kT_e)$, determined by the global energy balance

$$ewE = \int_0^\infty \varepsilon^{3/2} C_0^{en} \mathrm{d}\varepsilon = \frac{2\gamma N}{\pi^{1/2}(kT_e)^{3/2}} \int_0^\infty \times \left(\frac{2m_e}{M}\left(\varepsilon - \frac{3}{2}kT_0\right)\varepsilon\sigma_m^{en} + \sum_k U_k \varepsilon \sigma_k \right) \exp\left(-\frac{\varepsilon}{kT_e}\right) \mathrm{d}\varepsilon. \tag{36}$$

The C_0^{ee} term does not appear explicitly in this energy balance, due to (21).

Since the Maxwellian shape of the distribution function generally extends to higher energies than its shape without Coulomb collisions, the C_0^{ee} term tends to enhance the occurrence of high-threshold inelastic collisions. These collisions can then occur at much lower *E/N*. For the case of figure 1(a), the tail of F_0 exceeds the excitation and ionization thresholds if C_0^{ee} is included, but not otherwise. This Coulomb collision effect becomes important for ionization degrees $n_e/N > 10^{-5}$ and has been amply discussed in the literature [e.g. 5, 7, 9, 11, 19]. To illustrate this further, figure 3(b) shows the argon ionization rate coefficient

$$k_{iz} = \gamma \int_0^\infty \varepsilon \sigma_{iz}^{en} F_0 d\varepsilon \tag{37}$$

as function of *E/N* for different ionization degrees. For $n_e/N > 10^{-4}$, the *E/N* value needed to attain a certain ionization rate may be lowered by an order of magnitude with respect to the case without Coulomb collisions.

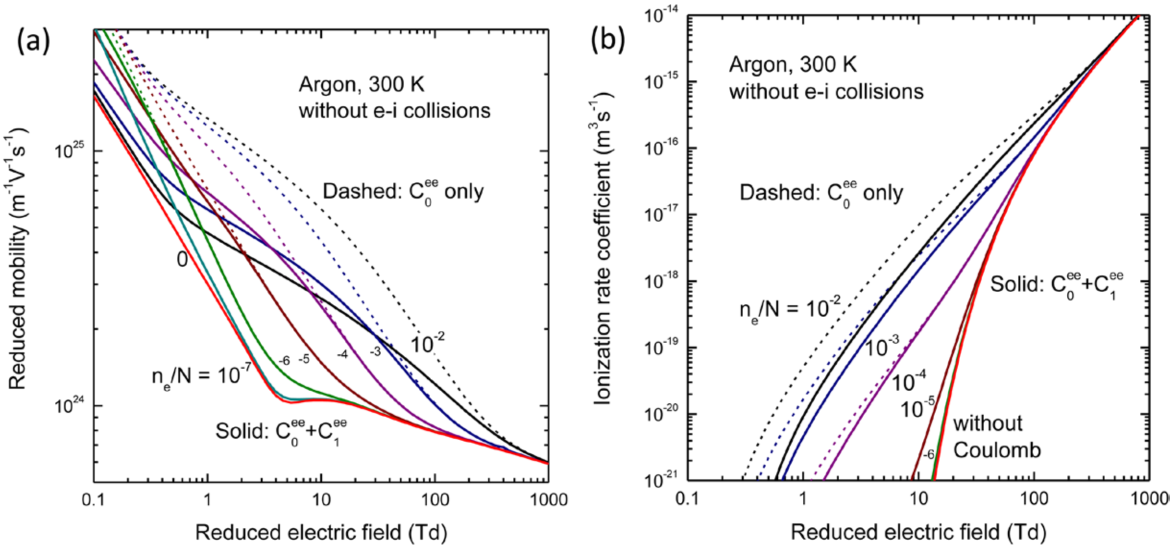

Figure 3. Effect of the electron-electron collision terms on the electron mobility (a) and the ionization rate coefficient (b) in argon as a function of reduced electric field for different ionization degrees. Electron-ion collisions are neglected.

3.2. Effect of electron-electron collisions on momentum transfer

The effect of electron-electron collisions on the shape of F_0 discussed in the previous section is relatively well known in low temperature plasma research. What is much less known is the influence of these collisions on F_1, the anisotropic part of the distribution function due to the electric field. Let us discuss this next, for the argon case of figure 1, neglecting electron-ion collisions.

Figure 2 shows the anisotropy, defined as the ratio of $|F_1|$ to F_0, for the same conditions as in figure 1. Without Coulomb collisions, combining equations (10) and (35) leads to

$$\frac{F_1}{F_0} = \frac{eE}{N\sigma_m^{en}}\frac{\partial \ln F_0}{\partial \varepsilon} = -\frac{6m_e}{M}\frac{N}{eE}\varepsilon\sigma_m^{en}, \tag{38}$$

which means that the anisotropy is small in the Ramsauer minimum of the momentum cross section near 0.2 eV, as can be seen in figure 2(a) for a reduced electric field of 1 Td (red curve). At first sight this may seem counterintuitive, but realize that in this case the electrons move ballistically back and forth through the Ramsauer minimum after being backscattered (against the electric force) at higher energy. At higher fields the minimum in the anisotropy is shifted and attenuated due to the influence of inelastic processes (figure 2(b), red curves).

Including the electron-electron C_0^{ee} term results in a behavior opposite to equation (38). Now, since the slope of F_0 is almost constant, equation (10) yields

$$\frac{F_1}{F_0} \approx -\frac{eE}{NkT_e\sigma_m^{en}}, \tag{39}$$

so the anisotropy becomes very large in the Ramsauer minimum (black curves). Locally the anisotropy can even exceed 1 which calls into question the two-term approximation (2). However, including also the C_1^{ee} term again changes this behavior. The shape of F_1 is now pushed toward that of the shifted Maxwellian function of equation (26):

$$\frac{F_1}{F_0} \approx -\frac{2w}{\gamma kT_e}\varepsilon^{1/2}. \tag{40}$$

In the log-log plot of figure 2 (blue curves) this shows up as a straight line with a slope of 1/2, independent of the shape of the electron-neutral cross section. In this last case, the magnitude of F_1 is determined by the mean velocity w from the global momentum balance:

$$-eE = \frac{2}{3}\int_0^\infty \varepsilon C_1^{en}\mathrm{d}\varepsilon = -\frac{8Nw}{3\pi^{1/2}\gamma(kT_e)^{5/2}} \times \int_0^\infty \varepsilon^2\sigma_m^{en}\exp(-\varepsilon/(kT_e))\mathrm{d}\varepsilon. \tag{41}$$

These different shapes of F_1 have important consequences for the electron mean velocity and transport coefficients. Figure 3(a) shows the electron mobility, defined as

$$\mu = -\frac{w}{E} = -\frac{\gamma}{3E}\int_0^\infty \varepsilon F_1\mathrm{d}\varepsilon, \tag{42}$$

calculated with BOLSIG+ as a function of *E/N* for different ionization degrees, with and without the different electron-electron collision terms C_0^{ee} and C_1^{ee}. It is clear from this figure that electron-electron collisions can enhance the electron mobility in argon by a large factor, but that this effect can be strongly overestimated if only C_0^{ee} in taken into account and C_1^{ee} is omitted. Using equations (39) and (41) for a Maxwellian F_0, the mobility without C_1^{ee}can be written as

$$\mu \approx \frac{2e\gamma}{3\pi^{1/2}NT_e^{5/2}}\int_0^\infty \frac{\varepsilon}{\sigma_m^{en}}\exp(-\varepsilon/T_e)\mathrm{d}\varepsilon, \tag{43}$$

whereas with C_1^{ee} we have

$$\mu \approx \frac{3\pi^{1/2}e\gamma T_e^{5/2}}{8N\int_0^\infty \sigma_m^{en}\varepsilon^2\exp(-\varepsilon/T_e)d\varepsilon}. \tag{44}$$

In the first case the inverse of the cross section is integrated, giving more weight to the Ramsauer minimum than the integral in the second case, which leads to a much larger mobility.

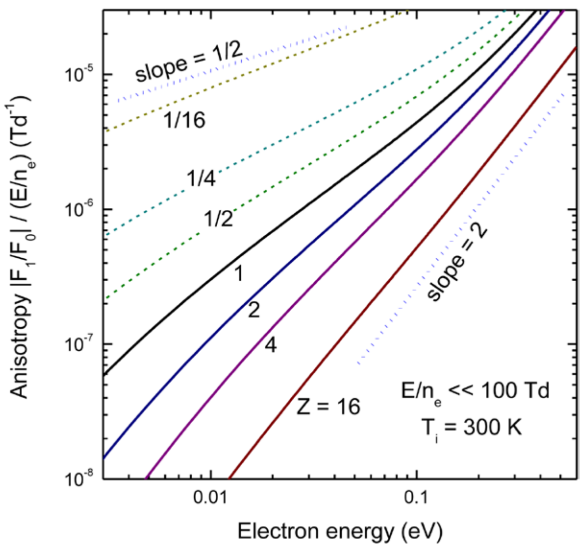

Figure 4. Anisotropy $|F_1/F_0|$ in the regime where electron-ion collisions dominate over electron-neutral collisions, for different values of the ion charge number.

That the C_1^{ee} term reduces the electron mobility is a general phenomenon which can be qualitatively understood as follows. If the momentum-transfer cross section σ_m^{en} varies strongly as a function of energy, then the electron-neutral collisions tend to deplete F_1 preferentially in certain energy regions where σ_m^{en} is largest. The C_1^{ee} term, diffusing electron momentum in energy space, repopulates F_1 in these collision-rich energy regions, so that the electron momentum is more efficiently transferred to the gas. This effect is stronger as the shape of the momentum-transfer cross section deviates more from $\sigma_m^{en} \propto 1/\varepsilon^{1/2}$. In the special case that $\sigma_m^{en} \propto 1/\varepsilon^{1/2}$, the electron-neutral collisions are not energy selective, the electron distribution function adopts the Maxwellian shapes (25) and (26), and equations (43) and (44) yield the same mobility; but for real gases this is never the case.

3.3. Momentum transfer in a Coulomb-collision dominated regime

For simplicity we neglected electron-ion collisions in the previous sections, but in most cases with a high ionization degree n_e/N, ions are also present, forming a quasi-neutral plasma. This leads to a delicate interplay between electron-electron collisions and electron-ion collisions, which we discuss next.

The electron-ion Coulomb collision terms (32) and (33) are basically similar to elastic electron-neutral terms (12) and (9). However, there is one important difference: the magnitude of these terms is linked with that of the electron-electron terms, so that the influence of electron-ion collisions grows along with that of electron-electron collisions as the ionization degree is increased. From the expressions in section 2 it appears that the competition between these two collision types depends essentially on the parameter:

$$Z = Z_i^2 n_i / n_e. \tag{45}$$

Table 1. Electron mobility in a regime dominated by Coulomb collisions. Comparison between BOLSIG+ and classical results from Spitzer and Härm [14].

Z	$\mu \cdot n_e$ (10^{18} V^{-1} s^{-1} m^{-1})	β BOLSIG+	β Spitzer
0	38.9/Z	0.295	$3\pi/32$
1	77.0	0.583	0.5816
2	45.3	0.687	0.6833
4	25.9	0.786	0.7849
16	7.62	0.924	0.9225
∞	132/Z	1.000	1

For a quasineutral plasma this reduces to the ion charge number Z_i and for an electropositive low-temperature plasma it is usually close to 1. However, generalizing (45) to the case of an electronegative plasma with two oppositely charged ions species, we have

$$Z = (Z_{i+}^2 n_{i+} + Z_{i-}^2 n_{i-})/n_e \approx 1 + 2n_{i-}/n_e, \tag{46}$$

which can become arbitrarily large as the ratio of the negative ion density n_{i-} to the electron density increases.

It is instructive to investigate the effect of the parameter Z in the limit of high ionization degree and weak electric field, where electron-neutral collisions are negligible. This corresponds in fact to the classical problem considered by Spitzer [14]. The electrons in this limit are in thermal equilibrium with the ions and acquire a Maxwellian energy distribution function F_0 with a temperature equal to the ion temperature, due to the C_0^{ei} term. However, the anisotropic part F_1 of the electron distribution function depends on the competition between the electron-electron C_1^{ee} term, trying to establish a shifted-Maxwellian function F_1 as given in equation (26), and the electron-ion C_1^{ei} term, trying to establish a different shape

$$F_1 = \frac{eE}{n_i \sigma_m^{ei}} \frac{\partial F_0}{\partial \varepsilon} = -\frac{16\pi\varepsilon_0^2 E}{e^3 \ln \Lambda Z n_e k T_e} \varepsilon^2 F_0. \tag{47}$$

This is illustrated by figure 4, which shows the anisotropy $|F_1/F_0|$ versus electron energy for different values of Z at a fixed ratio of $E/n_e = 1$ Td and an ion temperature of 300 K. For $Z \gg 1$, the effect of the C_1^{ee} term is negligible and the anisotropy follows expression (47) varying as ε^2, characterized by a slope of 2 in the log-log plot of the figure. For $Z \ll 1$, the C_1^{ee} term is dominant and the anisotropy shows the $\varepsilon^{1/2}$ variation of equation (26), i.e. slope 1/2 on the log-log plot. However, since Z cannot be smaller than 1 in a quasineutral plasma, this shifted-Maxwellian function can never be attained. At $Z = 1$ the shape of the anisotropy is somewhere in between the $\varepsilon^{1/2}$ and the ε^2 dependence.

These different shapes of the anisotropy affect the electron mobility, in a similar way as discussed in section 3.2 for the case of electron-neutral collisions. In the limit $Z \to \infty$, the mobility can be calculated analytically from equation (47):

$$\mu_\infty = -\frac{\gamma}{3} \int_0^\infty \frac{\varepsilon}{n_i \sigma_m^{ei}} \frac{\partial F_0}{\partial \varepsilon} d\varepsilon = \frac{64\pi^{1/2}\gamma\varepsilon_0^2 (kT_e)^{3/2}}{e^3 Z n_e \ln \Lambda} \tag{48}$$

For smaller Z, the electron mobility is reduced with respect to this analytical expression, due to the C_1^{ee} term repopulating F_1

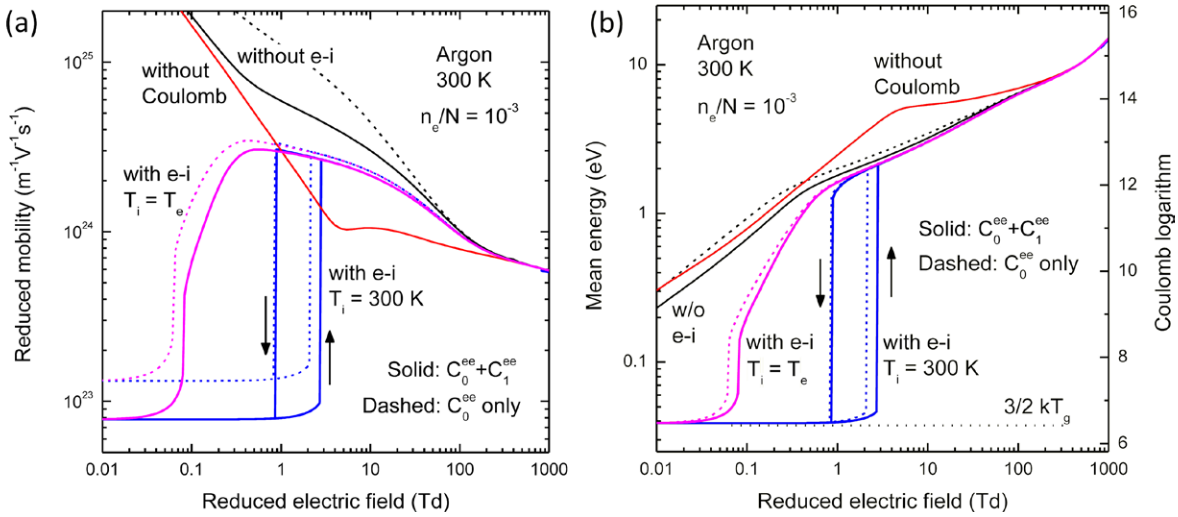

Figure 5. Effect of different Coulomb collision terms on the electron mobility (a) and mean energy (b) in argon gas with an ionization degree of 10^{-3}. The left axis of panel (b) indicates the Coulomb logarithm $\ln\Lambda$ for a plasma density of 10^{16} m^{-3}.

at low energy where electron-ion momentum transfer is most efficient:

$$\mu_S = \mu_\infty \beta = \frac{64\pi^{1/2}\gamma\varepsilon_0^2(kT_e)^{3/2}}{e^3 n_e \ln\Lambda}\frac{\beta(Z)}{Z}, \qquad \frac{3\pi}{32} < \beta(Z) < 1, \tag{49}$$

where β is a correction factor which is a unique function of Z. This corrected mobility corresponds to the well-known Spitzer conductivity for fully ionized plasmas, hence the subscript 'S' [Cohen50, Spitzer53]. The lower limit $\beta = 3\pi/32$ comes from a Maxwellian F_1 in the hypothetical limit $Z = 0$, but for real quasineutral plasmas β cannot become smaller than 0.58. Table 1 gives the electron mobility and the correction factor β for different Z as calculated with BOLSIG+ for the distribution functions of figure 4. These BOLSIG+ results are compared with values from the classic paper by Spitzer and Härm [14] (where they are labelled γ_E rather than β). The agreement is better than 0.3%.

3.4. Competition between electron-ion collisions and electron-neutral collisions

We will now discuss the role of electron-ion collisions at lower ionization degree, where electron-neutral collisions also play a role. Figure 5 shows the electron mobility and mean energy as a function of reduced electric field E/N at an ionization degree of $n_e/N = 10^{-3}$. (This is still a rather extreme ionization degree for a low-temperature plasma, chosen here for the sake of clarity of the figure, but the phenomena are qualitatively the same at much lower ionization degrees.) The ion mass is assumed equal to the neutral particle mass and $Z = 1$. The figure compares BOLSIG+ results obtained with and without different Coulomb collision terms.

If electron-ion collisions are included with a fixed ion temperature of 300 K (blue curves), then an abrupt change in the electron mobility and mean energy occurs at some critical electric field. Below this critical field the electrons are nearly in thermal equilibrium with the ions and have an energy distribution function F_0 close to a Maxwellian function (25) at 300 K; beyond this field the electrons are no longer thermalized and have a much higher energy and mobility. Within a certain reduced electric field range (0.8–3 Td in figure 5), these low- and high-energy solutions co-exist. The low-energy solution is found when gradually increasing the field from zero, the high-energy solution when decreasing the field from 1000 Td. Such bistable solutions can arise from the nonlinearity of the Coulomb collision terms. A related though different bistability, due to electron-electron collisions alone (without ions), is discussed in [20].

In the low-energy regime, the influence of electron-neutral collisions is completely negligible and the electron mobility obeys the Spitzer law (49). Therefore, a criterion for the existence of this regime can be found from the electron energy balance neglecting electron-neutral collisions:

$$\mu_S E^2 = \int_0^\infty \varepsilon^{3/2} C_0^{ei} d\varepsilon = \frac{32e}{\pi m_i \mu_\infty} k(T_e - T_i), \tag{50}$$

which yields the following equation for the electron temperature:

$$(T_e/T_i) - 1 = \beta \frac{m_i}{m_e}\left(\frac{16\pi\varepsilon_0^2 EkT_i}{e^3 Z \ln\Lambda n_e}\right)^2 (T_e/T_i)^3. \tag{51}$$

This equation has a stable solution at low electric field (small right-hand side <1/2) which ceases to exists if

$$E > E_{crit} \equiv \frac{e^3(m_e/m_i)^{1/2} Z \ln\Lambda}{24\pi\varepsilon_0^2(3\beta)^{1/2}}\frac{n_e}{kT_i}, \qquad T_e > \frac{3}{2}T_i. \tag{52}$$

Beyond this critical limit, there are not enough electron-ion collisions to limit the electron current and absorb the electron energy picked up by this current. Equation (52) is related to the critical electric field derived by Dreicer [26] as a threshold for electron and ion runaway in fully ionized plasmas. In our case of a weakly ionized low-temperature plasma, the electrons do not actually run away beyond this limit, but establish a new equilibrium due to

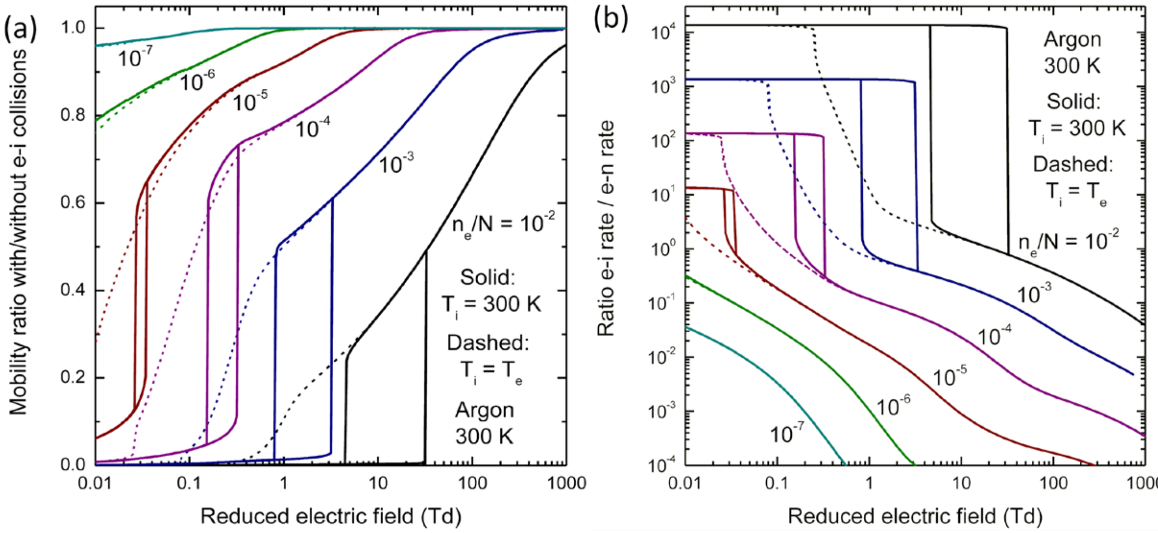

Figure 6. Effect of electron-ion collisions on the electron mobility and momentum transfer in argon at 300 K for different ionization degrees n_e/N: (a) ratio of the mobility obtained with electron-ion collisions to that without electron-ion collisions; (b) ratio of the electron-ion to electron-neutral momentum transfer frequencies as given in equation (53). The ions are assumed to have the same temperature as the neutral gas (solid lines) or as the electrons (dashed lines). The electron-electron collision terms are included for all curves.

electron-neutral collisions. For the $T_i = 300$ K case of figure 5, equation (52) predicts a regime transition at 3.0 Td which agrees very well with the numerical results. Note also the effect of the factor β on the results in figure 5. Omitting the C_1^{ee} term (dashed curves) results in $\beta = 1$ rather than $\beta = 0.58$, hence the Spitzer mobility is overestimated by a factor $1/0.58 = 1.72$ and the critical field underestimated by a factor $0.58^{1/2} = 0.76$.

However, it is questionable whether the ions maintain a fixed temperature of 300 K in a regime dominated by electron-ion collisions; they can in principle be heated up to the electron temperature. We will estimate the importance of this effect in the section 3.5. An upper limit is given by the case that the ion temperature becomes equal to the electron temperature, i.e. $kT_i = -1/(\partial \ln F_0/\partial\varepsilon)$, so that the C_0^{ei} term vanishes. This case is also shown in figure 5 (pink curves). There still exists a regime where the electrons (and ions) are thermalized near 300 K, but it occurs at much lower electric field and without bistability.

Above the critical field (52), electron-ion collisions no longer govern the electron balance, so the ion temperature has almost no effect, but they continue to cause significant electron momentum loss and decrease of the electron mobility. Their effect is nevertheless smaller than that of electron-electron collisions: the difference between the back and blue curves in figure 5 is smaller than that between the black and red curves, and it starts at lower electric field. In order to quantify separately the effect of electron-ion collisions on the electron mobility, figure 6(a) shows the ratio of the electron mobility μ calculated with the complete Coulomb collision terms, to the mobility μ' calculated with only the electron-electron terms, for different ionization degrees. In addition, figure 6(b) shows the ratio of the momentum transfer frequency for Coulomb collisions to that for electron-neutral collisions, calculated as

$$\frac{\nu_{ei}}{\nu_{en}} = \frac{n_i}{N}\frac{\int_0^\infty \varepsilon^{3/2}\sigma_m^{ei}F_1 d\varepsilon}{\int_0^\infty \varepsilon^{3/2}\sigma_m^{en}F_1 d\varepsilon} = \frac{n_e}{N}\frac{Ze^4\ln\Lambda}{16\pi\varepsilon_0^2}\frac{\int_0^\infty F_1/\varepsilon^{1/2} d\varepsilon}{\int_0^\infty \varepsilon^{3/2}\sigma_m^{en}F_1 d\varepsilon}. \quad (53)$$

This frequency ratio is related to the mobility ratio as $\mu/\mu' \approx 1/(1 + \nu_{ei}/\nu_{en})$ approximately. From these figures it appears that electron-ion momentum transfer can reduce the mobility by more than 10% over a wide parameter range.

Furthermore we note that in the low-energy thermalized regime, the Coulomb logarithm $\ln\Lambda$ (involved in the magnitude of the Coulomb collision terms) can be quite small and therefore sensitive to the value of the plasma density n_e and also to fundamental uncertainties in the underlying theory [16]. As $\ln\Lambda$ is a linear function of the logarithm of the electron mean energy, we have indicated it directly on the right axis of figure 5(b). At 300 K, $\ln\Lambda = 6.47$ for a plasma density of 10^{16} m^{-3} and is even smaller for higher plasma densities, decreasing by approximately 1.15 for every order of magnitude increase of n_e. Consequently the BOLSIG+ results for the thermalized regime depend not only on the ionization degree n_e/N but also significantly on the absolute value of n_e. This is illustrated by figure 7. When changing n_e by a factor 10 at constant n_e/N, the Spitzer mobility and critical electric field (52) change by approximately 20%, whereas the mobility above the critical field does not change more than 2%. In any case this effect remains much smaller than that of the ionization degree n_e/N and we will not further consider it here.

3.5. Effect of electron-ion collisions on ions

The usual assumption that the electrons in gas discharges interact with a fixed heavy particle background is questionable in case of an ion background having the same particle number

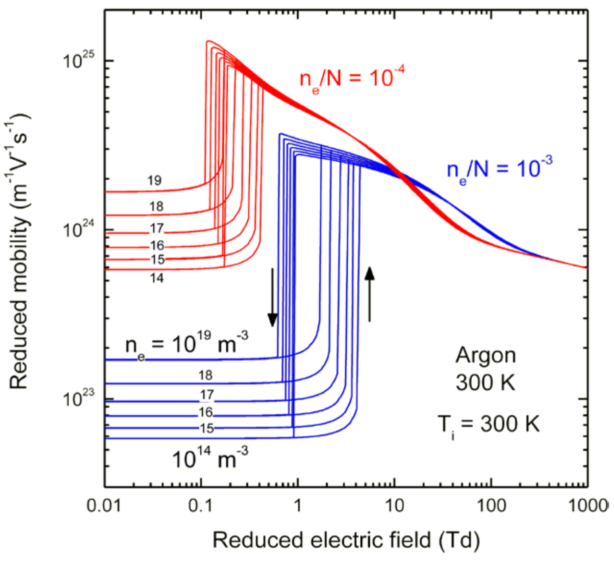

Figure 7. Effect of the absolute plasma density ($n_e = 10^{14}$, 10^{15}, 10^{16}, 10^{17}, 10^{18} and 10^{19} m^{-3}) for given ionization degrees $n_e/N = 10^{-3}$ (blue) and 10^{-4} (red), on the electron mobility in argon, calculated using the full Coulomb collision terms and a fixed ion temperature of 300 K.

density as the electrons themselves. All electron momentum and energy lost in electron-ion collisions is absorbed by the ions and affects their motion, depending on how efficiently they transfer it on to the neutral gas. To investigate this phenomenon, let us write macroscopic momentum balances for electrons and ions:

$$m_e\nu_{en}w_e + m_e\nu_{ei}(w_e - w_i) = -eE \tag{54}$$

$$m_i\nu_{in}w_i - m_e\nu_{ei}(w_e - w_i) = eE \tag{55}$$

where w_e and w_i are the electron and ion mean velocities and ν are macroscopic momentum transfer frequencies deduced from the Spitzer mobility and the electron and ion mobilities μ_{en} and μ_{in} in the absence of Coulomb collisions:

$$\nu_{ei} = e/(m_e\mu_S), \quad \nu_{en} = e/(m_e\mu_{en}), \quad \nu_{in} = e/(m_i\mu_{in}). \tag{56}$$

Combining these equations yields

$$w_i = -\frac{\mu_{in}}{\mu_{en}}w_e = \frac{\mu_S\mu_{in}E}{\mu_S + \mu_{en} + \mu_{in}} \approx \frac{\mu_{in}E}{1 + \mu_{en}/\mu_S}. \tag{57}$$

This means that electron-ion collisions do not modify the ratio of the electron and ion mean velocities: they reduce the ion mobility by the same factor as the electron mobility. Hence the fractional decrease of the electron mobility shown in figure 6 for different ionization degrees, also applies to the ion mobility. Moreover, since the ion mobility μ_{in} is in general much smaller than the electron mobility μ_{en}, the ion drift does almost not affect the electrons and can reasonably be neglected in electron Boltzmann calculations, even in a regime dominated by Coulomb collisions.

Next we estimate the ion temperature from the macroscopic ion energy balance

$$\nu_{in}\frac{3}{2}k(T_i - T_0) = eEw_i + \frac{32e}{\pi m_i\mu_\infty}k(T_e - T_i), \tag{58}$$

where the left-hand side is the ion energy transferred (per unit time) to neutrals in collisions, assuming equal masses $m_i = M$, and the right-hand side is the ion energy gained from the electric field and from electron-ion Coulomb collisions, as given in equation (50). Since the ion energy gained from the field is relatively small, in view of (57), it follows that

$$\frac{T_i - T_0}{T_e - T_i} \approx \frac{64\mu_{in}}{3\pi\mu_\infty}, \tag{59}$$

so the ion temperature is closer to the neutral gas temperature than to the electron temperature as long as the Spitzer mobility sufficiently exceeds the ion mobility. Substituting the expression for μ_∞, we find a simple criterion for the conditions where it is reasonable to assume a fixed ion temperature $T_i = T_0$ in the electron Boltzmann calculations:

$$\frac{n_e}{N} \ll \frac{3\pi^{3/2}\gamma\varepsilon_0^2(kT_0)^{3/2}}{e^3\ln\Lambda(\mu_{in}N)} \tag{60}$$

For argon, the ion mobility at low E/N without Coulomb collisions is $\mu_{in}N = 4.1 \times 10^{21}$ V^{-1}s^{-1} m^{-1} [30], so that $T_i = T_0 = 300$ K is a good assumption for $n_e/N \ll 5 \times 10^{-3}$. Consequently the results for $T_i = 300$ K in figures 5 and 6 are probably more relevant than those for $T_i = T_e$.

3.6. When are Coulomb collisions important?

In this section we present an overview of the conditions under which Coulomb collisions play a role in the simple dc discharge system described by our Boltzmann calculations.

The two main parameters controlling the importance of Coulomb collisions are the ionization degree n_e/N and the reduced electric field E/N. At sufficiently low n_e/N and/or high E/N, Coulomb collisions are completely negligible. Increasing n_e/N for a given E/N, or decreasing E/N for a given n_e/N, the influence of these collisions grows until the electron transport coefficients and rate coefficients start to be modified by a significant percentage. Figure 8 shows, in the 2D parameter space spanned by n_e/N and E/N, the lines where the electron mobility and the ionization rate coefficient in argon are changed by 3% when including the full Coulomb collision terms with respect to calculations without Coulomb collisions. These lines can be regarded as boundaries between regions where Coulomb collisions play a role, on the bottom-right side of the lines, and where they can be safely neglected, on the top-left side. The percentage of 3% is somewhat arbitrary; similar results are obtained for other small percentages (see figure 8(b)).

Figure 8(a) suggests that Coulomb collisions affect the ionization rate before affecting the mobility, as the boundary for ionization (dashed black line) lies much more to the top-left. However, the ionization rate coefficient k_{iz} is a very steep function of E/N (see figure 3(b)) so that any changes of k_{iz} can easily be compensated for by much smaller fractional changes of E/N, and this is in fact what tends to happen in real plasma discharges. To take this into account, we determined an alternative boundary for ionization (solid black line) from

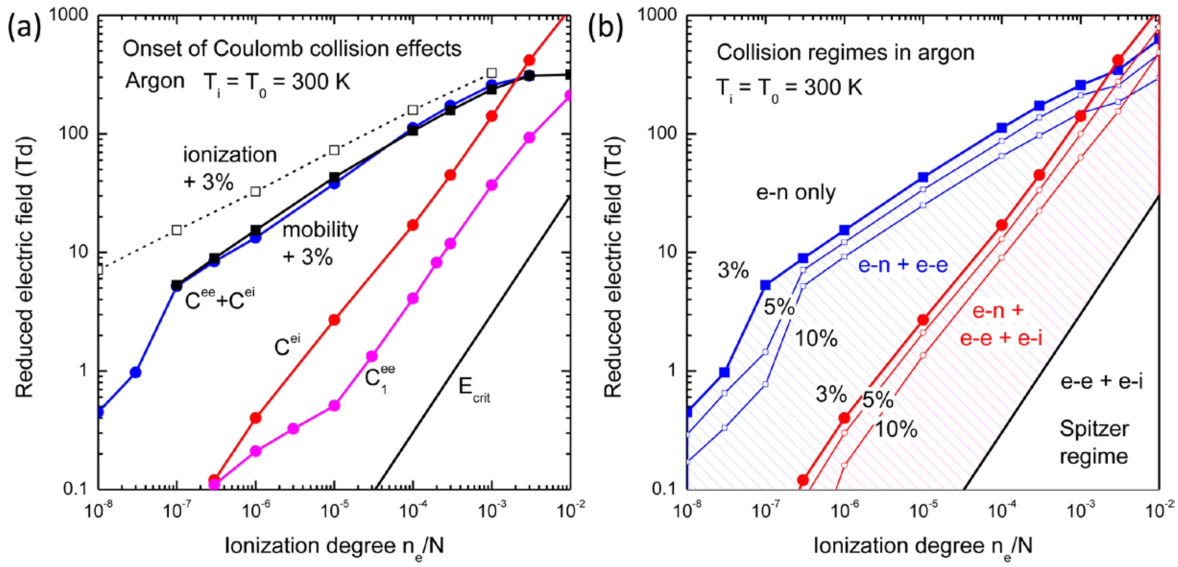

Figure 8. Subdivision of parameter space into regions where different Coulomb collision effects occur in argon gas at 300 K. See text for further explanation.

$$\frac{|\Delta k_{iz}|}{k_{iz}}\left(1+\left(\frac{\partial\ln(k_{iz})}{\partial\ln(E/N)}\right)^2\right)^{-1/2}=0.03. \qquad (61)$$

The left hand side of this equation corresponds to the perpendicular distance (shortest distance), between two curves of k_{iz} in a log-log plot versus E/N, as in figure 3(b), for a given midpoint located between these curves. We consider that this is a more relevant measure for the effect of Coulomb collisions on ionization than the fractional change $|\Delta k_{iz}|/k_{iz}$ at constant E/N (i.e. the vertical distance in figure 3(b)). The boundary from equation (61) is very close to the 3% boundary for mobility and other quantities such as the electron mean energy (not shown); hence the onset of Coulomb collision effects can be schematically characterized by a single boundary in parameter space, as shown in figure 8(b).

Figure 8 also shows separate boundaries for the importance of the electron-ion collision terms $C_{0,1}^{ei}$ and the anisotropic electron-electron term C_1^{ee}, defined by the lines where the mobility is changed by 3% when these terms are omitted, with respect to calculations including the full Coulomb collision terms. These lines are located at much larger n_e/N and weaker E/N than the overall effect boundaries, so there is a large region in parameter space where the effect of Coulomb collisions is primarily due to the C_0^{ee} electron-electron term. Finally, in the bottom right corner of the figure, we plotted the critical electric field from equation (52) as a boundary for the Spitzer regime where Coulomb collisions dominate over electron-neutral collisions and the electrons are near thermal equilibrium with the ions. In this region the Boltzmann calculation results are sensitive to the assumptions on the ion energy distribution function and on the plasma density through the Coulomb logarithm.

Another important parameter of the Boltzmann calculations is the discharge gas, involving a specific set of electron-neutral cross sections as well as a specific heavy particle mass. Figure 9 shows the 3% effect boundaries in the plane of n_e/N and E/N for the different noble gases and the molecular gas nitrogen. These results were obtained using the cross section sets from the SIGLO database [29] and assuming the ion mass equal to the neutral particle mass. All gases in figure 9 exhibit the effects and regimes described in the previous sections for argon. The conditions for which Coulomb collisions modify the mobility or ionization rate by more than 3% are remarkably similar for the heavier gases, whereas for the lighter gases helium and neon the 3% boundaries are shifted towards higher n_e/N and lower E/N. The Coulomb-dominated Spitzer regime shows the opposite trend and occurs more easily in the lighter gases, due to the larger mass ratio m_e/m_i resulting in more efficient electron-ion energy transfer.

The case of nitrogen gas is different from others in that it involves low-threshold inelastic processes of rotational and vibrational excitation which cause additional electron energy losses in the thermal energy range, dominating over the elastic energy losses. In order to be able to reproduce thermal equilibrium at low E/N in nitrogen, we took into account the superelastic inverse processes of all rovibrational excitations, assuming a 300 K Boltzmann distribution of the upper states. From the results in figure 9 it appears that these molecular processes do not qualitatively change the Coulomb collision effects. The only qualitative difference we observed between nitrogen and the noble gases is that in nitrogen there is no bistability as in figure 5; below the critical field (52) only the low-energy solution exists.

4. Conclusions

Full description of Coulomb collisions in two-term electron Boltzmann calculations involves separate collision terms acting on the isotropic part F_0 and on the anisotropic part F_1 of the electron distribution function, both for electron-electron collisions and for electron-ion collisions. In this paper we provide appropriate expressions for all of these four Coulomb

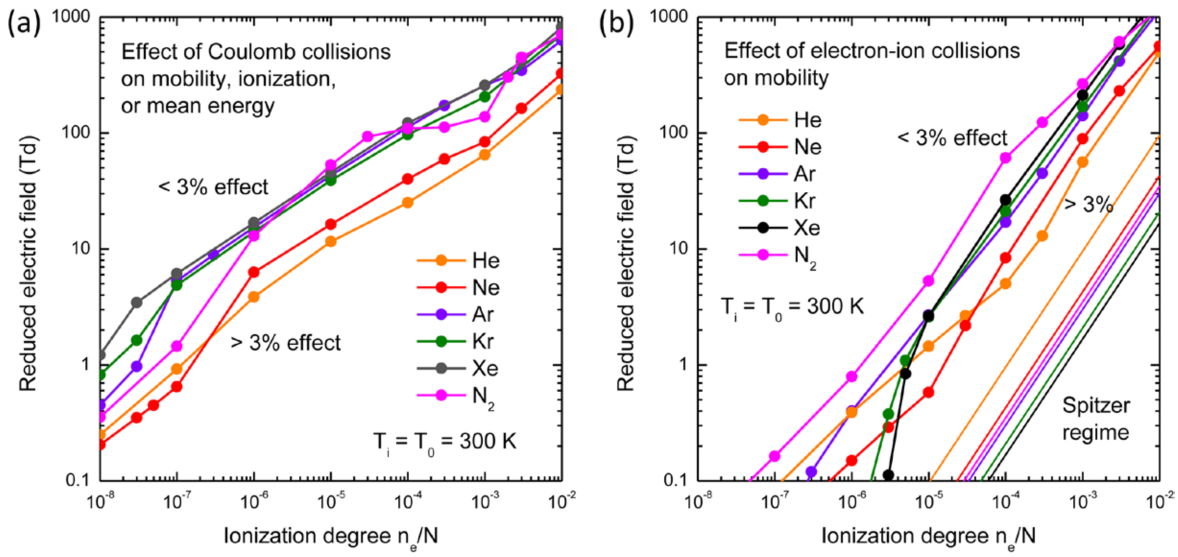

Figure 9. Regions in parameter space where the electron properties are affected by (a) full Coulomb collisions and (b) electron-ion collisions, in different gases at 300 K. See text for explanation.

collision terms, which we implemented in the freeware Boltzmann solver BOLSIG+ .

The four Coulomb collision terms induce distinct effects on the electron distribution function and transport coefficients, which can be of the same order of magnitude and interfere with one another. The BOLSIG+ results presented here demonstrate the following phenomena:

- Electron-electron collisions modify the shape of the electron energy distribution function F_0 such as to approach a Maxwellian distribution function. This in general enhances the high-energy tail of F_0 and consequently the rates of high-threshold inelastic processes such as ionization, as well as the electron mobility.
- Electron-electron collisions also smooth the shape of the anisotropy $|F_1/F_0|$ such as to enhance the efficiency of electron momentum transfer to the heavy particle background, which reduces the electron mobility, in opposition to F_0 effect above. This anisotropy effect is more important as the energy-dependence of the momentum-transfer cross-section differs more from $\propto 1/\varepsilon^{1/2}$; it is important in particular for electron-ion momentum transfer which varies as $1/\varepsilon^2$ and governs the Spitzer transport coefficients for highly ionized plasmas. Description of this effect requires the electron-electron collision term for F_1 which was neglected in previous works on gas discharges.
- Electron-ion collisions cause additional electron momentum and energy losses which induce an abrupt transition of regime at a certain critical electric field, related to the Dreicer field for electron-ion runaway known in the fusion plasma literature. Below this critical field the electron energy distribution function collapses to thermal energies and becomes strongly coupled to the ion distribution function. The electron transport coefficients in this regime are close to the Spitzer transport coefficients and much smaller than those without electron-ion collisions; the ion mobility is reduced by the same factor as the electron mobility.
- For electric fields stronger than the critical field, electron-ion collisions can considerably reduce the electron and ion mobilities (depending on the ionization degree) but their contribution to the electron energy losses is negligible.
- Under certain conditions (noble gas, high ionization degree, cold ions) a bistability occurs just below the critical field, involving two co-existing low- and high-energy solutions of the electron Boltzmann equation.

The importance of Coulomb collisions depends on the combination of ionization degree and reduced electric field, and to a lesser extent also on the discharge gas. It is therefore impossible to cite a single critical ionization degree for the onset of Coulomb collision effects in gas discharges. When increasing the ionization degree from zero at given reduced electric field, effects appear in the following order: first the ionization rate and mobility are increased by electron-electron collisions, then the electron and ion mobilities are decreased by electron-ion momentum transfer and the electron-electron anisotropy effect, and finally the electrons enter in thermal equilibrium with the ions and the Spitzer regime sets in. This latter regime can also show significant dependence on the heavy particle temperature and on the plasma density through the Coulomb logarithm. There is a large region in parameter space where only the electron-electron term for F_0 is of importance and the other Coulomb terms can be neglected, as in previous works.

The above conclusions hold for a dc discharge without space or time dependence in a fixed background gas as considered by most Boltzmann solvers used in low-temperature plasma modeling. However, the physical relevance of this simplified system for real discharge plasmas is questionable. In many real discharges, complex non-local or transient effects occur which could interfere with the Coulomb collision effects described here, and modify these conclusions.

It should furthermore be emphasized that the work presented in this paper is based on the traditional Coulomb collision theory with its inherent approximations concerning, for example, the binary collision picture and the definition of the

Coulomb logarithm $\ln\Lambda$, which could affect the accuracy and/or validity of the results, in particular for the low-energy thermalized regime where $\ln\Lambda$ is small (well below 10). It is generally assumed that the relative accuracy of the theory is of the order of $1/\ln\Lambda \approx 5$ to 20% [14, 16, 27] but this remains to be confirmed for the gas discharge conditions and parameters considered here. It is therefore desirable that future work be undertaken to check the results and conclusions from this work by alternative methods such as the particle simulations of [19].

Appendix

The classic paper by Rosenbluth *et al* [15] gives the following expression of the Fokker-Planck equation for particles of type *a* interacting with particles of type *b* through an inverse-square law of force:

$$\begin{aligned}\frac{v^2}{\Gamma}\left(\frac{\partial f_a}{\partial t}\right)_{col} &= -\frac{\partial}{\partial v}\left(v^2\frac{\partial h}{\partial v}f_a\right) - \frac{\partial}{\partial y}\left((1-y^2)\frac{\partial h}{\partial y}f_a\right)\\ &+ \frac{\partial^2}{\partial v^2}\left(\frac{1}{2}v^2\frac{\partial^2 g}{\partial v^2}f_a\right)\\ &+ \frac{\partial^2}{\partial y^2}\left\{\frac{1-y^2}{2v}\left(\frac{1-y^2}{v}\frac{\partial^2 g}{\partial y^2} + \frac{\partial g}{\partial v} - \frac{y}{v}\frac{\partial g}{\partial y}\right)f_a\right\}\\ &+ \frac{\partial^2}{\partial v\partial y}\left\{(1-y^2)\left(\frac{\partial^2 g}{\partial v\partial y} - \frac{1}{v}\frac{\partial g}{\partial y}\right)f_a\right\}\\ &+ \frac{\partial}{\partial v}\left\{\left(-\frac{1-y^2}{2v}\frac{\partial^2 g}{\partial y^2} - \frac{\partial g}{\partial v} + \frac{y}{v}\frac{\partial g}{\partial y}\right)f_a\right\}\\ &+ \frac{\partial}{\partial y}\left\{\left(\frac{y(1-y^2)}{2v^2}\frac{\partial^2 g}{\partial y^2} + \frac{(1-y^2)}{v}\frac{\partial^2 g}{\partial v\partial y}\right.\right.\\ &\left.\left.+ \frac{y}{v}\frac{\partial g}{\partial v} - \frac{1}{v^2}\frac{\partial g}{\partial y}\right)f_a\right\} \qquad (A.1)\end{aligned}$$

where Γ is defined as in equation (17), $y = \cos\theta$ is the cosine of the velocity angle with respect to an axis of symmetry, f is the particle distribution function in velocity space (v,y) and the variables g and h are integral functions over the background particle distribution function f_b, known as Rosenbluth potentials. Approximating the distribution functions by the usual two-term spherical harmonics expansion, we have

$$\begin{aligned}f_a(v,y) &= f_0(v) + f_1(v)y, \quad f_b(v,y) = f_0^b(v) + f_1^b(v)y\\ g(v,y) &= g_0(v) + g_1(v)y, \quad h(v,y) = h_0(v) + h_1(v)y \qquad (A.2)\end{aligned}$$

with, from Rosebluth's equations (45)–(46) and (40)–(41),

$$\begin{aligned}g_0(v) &= vP_0^2(v) + Q_0^3(v) + \frac{1}{3v}P_0^4(v) + \frac{1}{3}v^2Q_0^1(v)\\ g_1(v) &= -\frac{1}{3}P_1^3(v) - \frac{1}{3}vQ_1^2(v) + \frac{1}{15v^2}P_1^5(v) + \frac{1}{15}v^3Q_1^0(v)\\ h_0(v) &= (1+\alpha)\left(\frac{1}{v}P_0^2(v) + Q_0^1(v)\right)\\ h_1(v) &= (1+\alpha)\left(\frac{1}{3v^2}P_1^3(v) + \frac{1}{3}vQ_1^0(v)\right) \qquad (A.3)\end{aligned}$$

where $\alpha = m_a/m_b$ is the particle mass ratio and

$$\begin{aligned}P_{0,1}^m(v) &= 4\pi\int_0^v (v')^m f_{0,1}^b(v')dv',\\ Q_{0,1}^m(v) &= 4\pi\int_v^\infty (v')^m f_{0,1}^b(v')dv'. \qquad (A.4)\end{aligned}$$

We substitute the two-term distribution functions (A.2) and potentials (A.3) in the Fokker-Planck equation (A.1), neglect all second-order terms in $f_1 \cdot f_1$, multiply by 1 and y, respectively, integrate over y from $y = -1$ to $+1$ and rearrange the terms:

$$\frac{v^2}{\Gamma}\left(\frac{\partial f_0}{\partial t}\right)_{col} = \frac{\partial}{\partial v}\left(\alpha P_0^2 f_0 + \frac{1}{3v}(P_0^4 + v^3Q_0^1)\frac{\partial f_0}{\partial v}\right) \qquad (A.5)$$

$$\begin{aligned}\frac{v^2}{\Gamma}\left(\frac{\partial f_1}{\partial t}\right)_{col} &= -\frac{1+\alpha}{v}(P_0^2 f_1 - v^2Q_1^0 f_0)\\ &+ \frac{1}{v}\frac{\partial}{\partial v}\left(\alpha vP_0^2 f_1 + \frac{1}{3}v(P_0^4 + v^3Q_0^1)\frac{\partial(f_1/v)}{\partial v}\right.\\ &\left.+ \frac{1}{3}((2\alpha-1)P_1^3 - (1+\alpha)v^3Q_1^0)f_0 + \frac{1}{5v}(P_1^5 + v^5Q_1^0)\frac{\partial f_0}{\partial v}\right) \qquad (A.6)\end{aligned}$$

These equations are equivalent to those given in section 7.6 of [16]. Applying the variable change $v \to \varepsilon$ and generalizing for an arbitrary background charge number Z_b, we obtain equations (13) and (14), where

$$F_{0,1}^b(u) = \frac{2\pi\gamma^3\alpha^{3/2}}{n_b}f_{0,1}^b(\gamma(\alpha u)^{1/2}) \qquad (A.7)$$

and

$$A_{0,1}^m(\varepsilon) = \frac{P_{0,1}^{m+1}(\gamma\varepsilon^{1/2})}{n_b\gamma^{m-1}}, \quad B_{0,1}^m(\varepsilon) = \frac{Q_{0,1}^{m+1}(\gamma\varepsilon^{1/2})}{n_b\gamma^{m-1}}. \qquad (A.8)$$

References

[1] Frost L S and Phelps A V 1962 Rotational excitation and momentum transfer cross sections for electron in H_2 and N_2 from transport coefficients *Phys. Rev.* **127** 1621–33
[2] Crompton R W, Gibson D K and McIntosh A I 1969 Rotational cross sections in hydrogen *Aust. J. Phys.* **22** 715–32
[3] Petrovic Z Lj *et al* 2009 Swarm parameters and their application in plasma modelling *J. Phys. D: Appl. Phys.* **42** 194002
[4] Morgan W L and Penetrante B M 1990 ELENDIF: a time dependent Boltzmann solver for partially ionized plasmas *Comput. Phys. Commun.* **58** 127–52
[5] Hagelaar G J M and Pitchford L C 2015 Solving the Boltzmann equation to obtain electron transport coefficients and rate coefficients for fluid models *Plasma Sources Sci. Technol.* **14** 722–33; freeware code BOLSIG+ version 07/2015 available from www.bolsig.laplace.univ-tlse.fr (2015)

[6] Comsol Multiphysics, Plasma Module, www.comsol.com
[7] Dreicer H 1960 Electron velocity distributions in a partially ionized gas *Phys. Rev.* **117** 343–54
[8] Rockwood S D 1973 Elastic and inelastic cross sections for electron-Hg scaterring from Hg transport data *Phys. Rev.* A **8** 2348–58
[9] Yousfi M, Zissis G, Alkaa A and Damelincourt J J 1990 Boltzmann-equation analysis of electron kinetics in a positive column of low-pressure Hg-rare-gas discharges *Phys. Rev.* A **42** 978–88
[10] Dyatko N, Kochetov I V and Napartovich A P 1993 Electron energy distribution function in decaying nitrogen plasmas *J. Phys. D: Appl. Phys.* **26** 418–23
[11] Colonna G, Gorse C, Capitelli M, Winkler R and Wilhelm J 1993 The influence of electron-electron collisions on electron energy distribution function in N_2 post discharge *Chem. Phys. Lett.* **213** 5–9
[12] Loffhagen D 2005 Impact of electron-electron collisions on the spatial electron relaxation in non-isothermal plasmas *Plasma Chem. Plasma Process.* **25** 519–38
[13] Cohen R S, Spitzer L and Routly P 1950 The electrical conductivity of an ionized gas *Phys. Rev.* **80** 230–8
[14] Spitzer L and Härm R 1953 Transport Phenomena in a completely ionized gas *Phys. Rev.* **89** 977–81
[15] Rosenbluth M N, MacDonald W M and Judd D L 1957 Fokker-Planck equation for an inverse-square force *Phys. Rev.* **107** 1–6
[16] Shkarofsky I P, Johnston T W and Bachynski M P 1966 *The Particle Kinetics of Plasmas* (Reading, MA: Addison-Wesley)
[17] Hirshman S P 1977 Transport of a multiple-ion species plasma in the Pfirsch-Schlüter regime *Phys. Fluids* **20** 589
[18] Karney C C F 1986 Fokker-Planck and quasilinear codes *Comp. Phys. Rep.* **4** 183–244
[19] Donko Z 2014 First principles calculation of the effect of Coulomb collisions in partially ionized gases *Phys. Plasmas* **21** 043504
[20] Dyatko N and Donko Z 2015 Bistable solutions for the electron energy distribution function in electron swarms in xenon: a comparison between the results of first-principles particle simulations and conventional Botlzmann equation analysis *Plasma Sources Sci. Technol.* **24** 045002
[21] Lieberman M A and Lichtenberg A J 2005 *Principles of Plasma Discharges and Materials Processing* 2nd edn (Hoboken, NJ: Wiley)
[22] Holstein T 1946 Energy distribution of electrons in high frequency gas discharges *Phys. Rev.* **70** 367–84
[23] Allis W P 1956 Motion of ions and electrons *Handbuch der Physik* ed S Flugge, vol 21 (Berlin: Springer) pp 383–444
[24] Kumar K, Skullerud H R and Robson R E 1980 Kinetic theory of charged particle swarms in neutral gases *Aust. J. Phys.* **33** 343–448
[25] Chandrasekhar S 1943 Stochastic problems in physics and astronomy *Rev. Mod. Phys.* **15** 1–89
[26] Dreicer H 1959 Electron and ion runaway in a fully ionized gas. I *Phys. Rev.* **115** 238–49
[27] Huba J D 2012 *NRL Plasma Formulary* GEC 2012 edition (Washington, DC: Naval Research Laboratory)
[28] Baalrud S D and Daligault J 2013 Effective potential theory for transport coefficients across coupling regimes *Phys. Rev. Lett.* **110** 235001
[29] Siglo database 2013 www.lxcat.net retrieved on 4 June 2013
[30] Ellis H W, Pai R Y, McDaniel E W, Mason A and Viehland L A 1976 Transport properties of gaseous ions over a wide energy range *At. Data Nucl. Data Tables* **17** 177–210